9.1 Classification:

The minerals required in human nutrition can be classified into two main groups.

Macro elements

Macro elements like sodium and potassium are required in large amount (>100mg/day) and present in large quantities in the body.

Micro elements

Micro elements like iron and iodine are required in small quantities (<100mg/day) and are present in small amount in tissues and body fluids.

9.2 Macro elements:

Some of the macro elements are calcium, phosphorus, sodium, potassium, chloride, magnesium and sulphur.

9.2.1Calcium

Calcium (Ca) is the most abundant inorganic mineral in the body. About 99% of it exists in the bone and teeth and 1% in soft tissues and extracellular fluid.
Functions:

- Calcium plays a role in blood coagulation by producing substances for thromboplastic activity of blood.
- Calcium is essential for bone and teeth formation.
- Calcium ions are necessary for nerve transmission and muscle contraction.
- Normal excitability of heart is calcium dependent.
- Calcium is involved in the process of mitosis.
- It also acts as cofactor for certain enzymes like succinate dehydrogenase.
- Calcium plays a role as secondary messenger in hormone action.
- Calcium is involved in the process of membrane fusion in endocytosis and exocytosis.
- It involves in maintenance of plasma membrane potential.
- Calcium binds trypsin near its active site to prevent autodigestion of trypsin.

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